

Mid Term Examination, Odd Semester 2023-24

B.Tech (CSE), III yr, V Semester

BCSE 0101: Digital Image Processing

Maximum Marks: 30

Time: 2 Hours

Course Outcome

- CO1- Understand mathematical formulation of an image, its processing steps and relationship between image pixels.
 CO2- Apply Image enhancement using intensity transformations and spatial filtering.
 CO3- Analyze image enhancement for frequency domain using Fourier transform.
 CO4- Formulate region of interest through morphological operations.
 CO5- Evaluate strongly co-related regions obtained through Segmentation using discontinuity and homogeneity-based segmentation techniques
 CO6- Describe an object of an image using Shape Number and Boundary descriptors.

Instruction for students:

- Use of Scientific Calculator is allowed.
- Clearly mention giving proper reason, if any assumption is made.
- No over writing is allowed – strike off and write it again where required.

Section – A

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	<u>Attempt All of the following.</u> a. Assume that you are given grey scale image with pixel intensities between [37 ... 209]. What point operation is necessary to map the pixels to the [0 ... 255] range? b. Suppose there is a multispectral image of size 100 x 100. This image has 4 bands and each band is using 256 gray levels. Find the size of the image in bits. c. If $X = \begin{bmatrix} 2 & 3 & 4 & 3 & 4 & 5 & 6 \end{bmatrix}$ and $w = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ what will be the result of Median filtering? Use Zero padding if required.	3	1, 2	A	C
2	<u>Attempt All of the following.</u> a. A tourist visited Raskhan Samadhi but was saddened when he saw the photo (given on right) that he had clicked was blurred. How can you help improve the quality of the image? 	3	3, 1	U	F

	b. The distance measure $D(p, q) = a - r + b - c$, where the coordinates of point p are (a, b) and that of q are (r, c), is known as _____ c. Consider the image shown below. Find the shortest m - adjacent path from pixel p to q. $V = \{1, 2\}$. Draw the image in your answer sheet & mark the path. <table border="1"> <tr><td>3</td><td>1</td><td>2</td><td>1 (q)</td></tr> <tr><td>2</td><td>2</td><td>0</td><td>2</td></tr> <tr><td>1</td><td>2</td><td>1</td><td>1</td></tr> <tr><td>(p) 1</td><td>0</td><td>1</td><td>2</td></tr> </table>	3	1	2	1 (q)	2	2	0	2	1	2	1	1	(p) 1	0	1	2																																																																						
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(p) 1	0	1	2																																																																																				
3	<u>Attempt any one</u> a. Consider the following image strip of 19 pixels. Compute its First order derivative. <table border="1"> <tr> <td>6</td><td>6</td><td>6</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td> </tr> </table> Your answer should be in a tabular format having 2 rows as shown below. First row should show original signal; second row should have values of the First Order Derivative. <table border="1"> <tr> <td>6</td><td>6</td><td>6</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td> </tr> <tr> <td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td> </tr> </table> b. Compute the convolution of the Laplacian kernel L_8 with the image given below. Use border values to extend the image. <table border="1"> <tr><td>100</td><td>100</td><td>10</td><td>10</td><td>10</td></tr> <tr><td>100</td><td>100</td><td>10</td><td>10</td><td>10</td></tr> <tr><td>100</td><td>100</td><td>10</td><td>10</td><td>10</td></tr> <tr><td>100</td><td>100</td><td>10</td><td>10</td><td>10</td></tr> <tr><td>100</td><td>100</td><td>10</td><td>10</td><td>10</td></tr> </table>	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6																				100	100	10	10	10	100	100	10	10	10	100	100	10	10	10	100	100	10	10	10	100	100	10	10	10	3	2	A	C
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4	<u>Attempt All of the following</u> a. Consider the following two images $f_1 = \begin{pmatrix} 100 & 150 & 70 \\ 50 & 150 & 175 \\ 200 & 50 & 150 \end{pmatrix} \text{ and } f_2 = \begin{pmatrix} 50 & 50 & 25 \\ 45 & 55 & 50 \\ 50 & 50 & 75 \end{pmatrix}$ Find $f_1 + f_2$ and $f_1 - f_2$. b. Which of the following is true for Power Law Transformation? Write the complete true statement in your answer sheet.	3	2	A	C																																																																																		

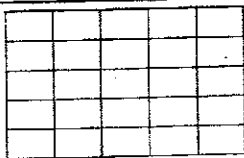
	• $\gamma > 1$: Compresses dark values; Expands bright values • $\gamma > 1$: Expands dark values; Compresses bright values • $\gamma < 1$: Compresses dark values; Expands bright values • $\gamma = 1$: Expands dark values; Compresses bright values																																																										
5	A and B are two 3 - bit images. Histogram Equalization was done on both the images and following results were obtained. <table border="1"> <tr> <td>I/P Gray Level of Image A</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> </tr> <tr> <td>O/P Gray Level of Image A</td> <td>1</td> <td>3</td> <td>5</td> <td>6</td> <td>6</td> <td>7</td> <td>7</td> <td>7</td> </tr> </table> <table border="1"> <tr> <td>I/P Gray Level of Image B</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> </tr> <tr> <td>O/P Gray Level of Image B</td> <td>0</td> <td>0</td> <td>0</td> <td>1</td> <td>2</td> <td>5</td> <td>6</td> <td>7</td> </tr> </table> We want to match the histogram of image A to that of image B. What will be the result of Histogram matching? Give your answer in the following format. <table border="1"> <tr> <td>I/P Gray Level of Image A</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> </tr> <tr> <td>O/P Gray Level of Image A after matching</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>	I/P Gray Level of Image A	0	1	2	3	4	5	6	7	O/P Gray Level of Image A	1	3	5	6	6	7	7	7	I/P Gray Level of Image B	0	1	2	3	4	5	6	7	O/P Gray Level of Image B	0	0	0	1	2	5	6	7	I/P Gray Level of Image A	0	1	2	3	4	5	6	7	O/P Gray Level of Image A after matching									3	2	An	P
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I/P Gray Level of Image B	0	1	2	3	4	5	6	7																																																			
O/P Gray Level of Image B	0	0	0	1	2	5	6	7																																																			
I/P Gray Level of Image A	0	1	2	3	4	5	6	7																																																			
O/P Gray Level of Image A after matching																																																											

Section - B

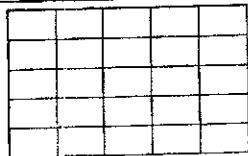
Attempt All Questions

5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL																									
6	<u>Attempt any ONE</u>	5	2	A	C																									
	a. Perform Gray Level slicing on the following 3 bit image. Let $A = 3$ & $B = 5$. Give the modified image with and without background transformations in the format given below. Also plot the graph for both types of transformations.																													
	<table><tr><td>2</td><td>2</td><td>1</td><td>1</td><td>2</td></tr><tr><td>2</td><td>3</td><td>4</td><td>5</td><td>2</td></tr><tr><td>6</td><td>7</td><td>6</td><td>7</td><td>2</td></tr><tr><td>2</td><td>6</td><td>5</td><td>6</td><td>1</td></tr><tr><td>1</td><td>0</td><td>3</td><td>3</td><td>1</td></tr></table>	2	2	1	1	2	2	3	4	5	2	6	7	6	7	2	2	6	5	6	1	1	0	3	3	1				
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2	3	4	5	2																										
6	7	6	7	2																										
2	6	5	6	1																										
1	0	3	3	1																										



Modified image without Background



Modified image with Background

- b. Find the values of logical operation XOR and set operation $A - B$ for the binary images A and B. Assume 1 to be foreground pixels.

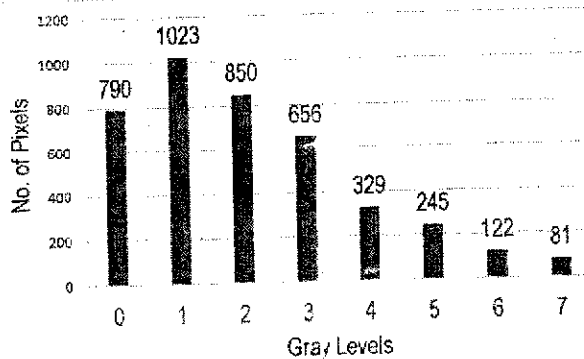
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	0
0	0	0	0	1	1	1	0
0	0	0	0	1	1	1	0
0	0	0	0	0	0	0	0

A

0	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0
0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

B

Following is the Histogram of a 3 bit image. Plot the histogram of the Equalized image. Show all the calculations in tabular format.



The following image got corrupted while it was being captured. Find its Fourier Transform so that it can be improved.

0	1	2	1
1	2	3	2
2	3	4	3
1	2	3	2

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Twiddle matrix